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Schweizer Norm
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SIA 260:2013 Construction

505 260

Replaces Code SIA 260:2003

Grundlagen der Projektierung von Tragwerken
Bases pour l'élaboration des projets de structures porteuses
Basi per la progettazione di strutture portanti

Basis of Structural Design

260

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FOREWORD

The present code SIA 260 is directed towards design engineers. Also addressed are owners and those involved in site supervision and the execution of construction works.

SIA 260 is part of the Swiss structural codes. It follows in general the European Standard EN 1990 *Basis of Structural Design*, while at the same time integrating the principles laid down in the code SIA 160 (1989).

The Swiss structural codes comprise the following:
- Code SIA 260 Basis of structural design
- Code SIA 261 Actions on structures
- Code SIA 262 Concrete structures
- Code SIA 263 Steel structures
- Code SIA 264 Composite steel and concrete structures
- Code SIA 265 Timber structures
- Code SIA 266 Masonry
- Code SIA 267 Geotechnical design.

The principles and the procedure for preservation of existing structures are covered by codes SIA 269 and SIA 269/1 to 269/8. New section 0.1.5 points this out explicitly.

With SIA 260 the technical terms “basis of design”, “service criteria agreement” and “structural concept” have been introduced for the first time.

Various terms have been used previously in Switzerland but were defined for the first time within the framework of the Swiss structural codes, e.g. “capacity design”, “conceptual design”, “configuration”, “deformation capacity”, “ductility”, “integration” and “observational method”.

With the partial revision, the term “defect” has been modified and the term “service life” has been introduced, to be compatible with the series of codes SIA 269. Furthermore, the term “non load-bearing components” has been removed from the structural codes, because each structural member carries at least itself and normally also wind and seismic forces.

Commission SIA 260

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0 SCOPE

0.1 Limitations

0.1.1 The code SIA 260 establishes the basic principles for the design of structures. Aspects dealing with execution, use and preservation are only considered in so far as they are relevant to design.

0.1.2 SIA 260 is valid in combination with the codes SIA 261 to 267. For actions and applications outside the scope of these codes the basic design principles shall be applied in the general sense of SIA 260.

0.1.3 The principles established in SIA 260 apply to all types of structure. The extent and content of the investigations shall be adapted to the importance of the construction and the complexity of the problems encountered.

0.1.4 The principles established in SIA 260 also apply to construction states and to temporary structures.

0.1.5 Preservation of existing structures is regulated in the codes SIA 269 and SIA 269/1 to 269/8.

0.2 Prerequisites

0.2.1 Design and execution shall be carried out under the supervision of appropriately qualified engineers. Their knowledge and experience in materials technology, structural analysis, construction technology, civil engineering practice and other relevant areas must correspond to the importance and the complexity of the construction project.

0.2.2 During the design, execution, use and preservation, suitable quality assurance measures shall be planned and implemented.

0.2.3 The structures shall be used as agreed and properly monitored and maintained according to the monitoring and the maintenance plan.

0.3 Deviations

0.3.1 Exceptions to the present code are permissible, provided they are well founded theoretically or experimentally, or justified by new developments and new knowledge.

0.3.2 Deviations from the code shall be clearly documented in the documents for construction together with the reasons for such deviation.

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1 TERMINOLOGY

1.1 Technical terms

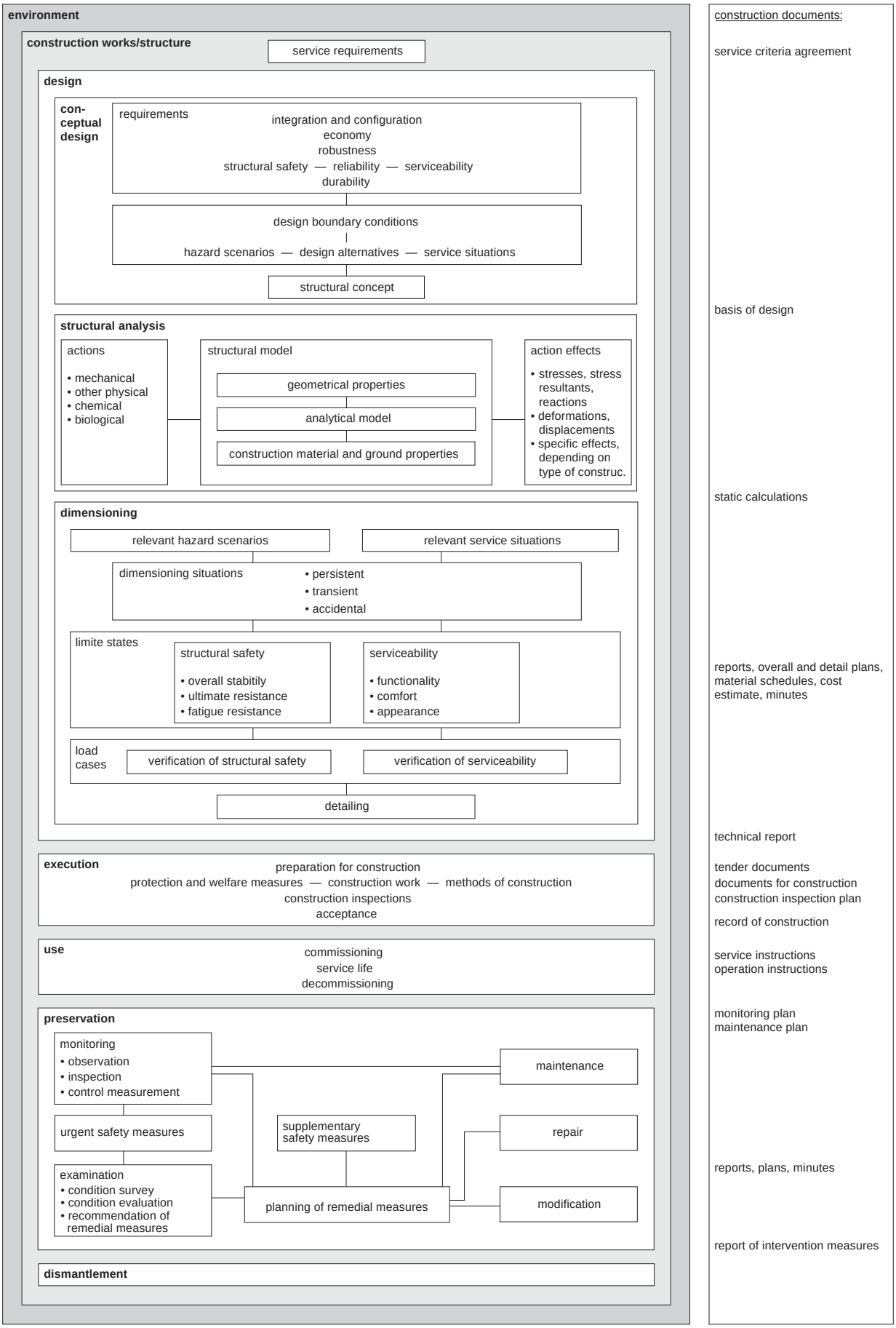
In the present code and in the codes SIA 261 to 267 the general technical terms defined below are used. Their context follows from Figure 1. Specific technical terms are defined in the codes SIA 261 to 267.

Acceptance *Abnahme* *réception* *collaudo*	The process of handing over the construction works or a part of it to the owner.
Accidental action *Aussergewöhnliche Einwirkung* *action accidentelle* *azione accidentale*	Action with low probability of occurrence, usually of short duration and considerable effect.
Accidental dimensioning situation *Aussergewöhnliche Bemessungssituation* *situation de dimensionnement accidentelle* *situazione di dimensionamento accidentale*	Dimensioning situation taking into account accidental conditions for the structure.
Accompanying action *Begleiteinwirkung* *action concomitante* *azione concomitante*	An action accompanying the leading action.
Accompanying hazard *Begleitgefahr* *danger concomitante* *rischio concomitanto*	Affiliated hazard in a hazard scenario.
Actions *Einwirkungen* *actions* *azioni*	Mechanical (loads, forces), other physical (temperature, humidity), chemical (salts, acids and alkaline substances, organic compounds) and biological (bacteria, insects, fungi, algae) actions on the structure resulting from the execution and use as well as from environmental influences.
Analytical model *Berechnungsmodell* *modèle d'analyse* *modello di calcolo*	Relating physical quantities (e.g. force and deformation quantities) through corresponding relationships (e.g. equilibrium conditions, constitutive relationships and kinematic conditions).
Basis of design *Projektbasis* *base du projet* *base del progetto*	Technical description of the implementation of the service criteria agreement.
Capacity design *Kapazitätsbemessung* *dimensionnement en capacité* *dimensionamento per capacità*	Method of seismic design with appropriately defined areas of plastic deformation exhibiting adequate ductility, together with other areas of the structure that are provided with increased ultimate resistance to ensure elastic behaviour.
Characteristic value *Charakteristischer Wert* *valeur caractéristique* *valore caratteristico*	The value of an action, a geometrical property or a construction material or ground property (mean value, upper or lower value), usually determined on a statistical basis, in some cases nominal value or prudent estimate.

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Figure 1: Relationships between different design elements



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Commissioning *Inbetriebnahme* *mise en service* *messa in servizio*	Start of agreed use.
Conceptual design *Entwurf* *conceptual design de projet* *progetto strutturale*	All the activities and developments leading from the service criteria to the structural concept and their result.
Condition evaluation *Zustandsbeurteilung* *évaluation de l'état* *valutazione dello stato*	Summary analysis and evaluation of the information on the current condition of the construction and the previous development of the condition, combined with a prediction of the further development of the condition and its consequences during the remaining service life.
Condition survey *Zustandserfassung* *relevé de l'état* *rilievo dello stato*	Acquiring information on the current condition of the construction and the previous development of the condition with the aim of recognising defects, deteriorations and deterioration mechanisms.
Configuration *Gestaltung* *configuration* *configurazione*	Creation of an aesthetic expression by means of spatial arrangement, shaping and choice of construction materials.
Construction documents *Bauwerksakten* *dossier de l'ouvrage* *documenti di costruzione*	Documents of the specific construction.
Construction inspection plan *Kontrollplan* *programme de contrôle* *piano di controllo*	Specifying the type, extent, execution and point in time of construction inspections, including information on quality requirements and admissible deviations as well as resolving questions of responsibilities and information flow.
Construction inspections *Ausführungskontrollen* *contrôles d'exécution* *controlli d'esecuzione*	Checking that the design specifications are implemented correctly during execution.
Construction materials *Baustoffe* *matériaux de construction* *materiali da costruzione*	Materials used for a construction.
Construction product *Bauprodukt* *produit de construction* *prodotto per costruzioni*	Product that is manufactured for permanent installation in a construction works.
Construction work *Bauausführung* *travaux d'exécution* *esecuzione dei lavori*	Carrying out the construction according to contract.
Construction works *Bauwerk* *ouvrage* *costruzione*	Product of construction work.

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Control measurement *Kontrollmessung* *contrôle par mesures* *misurazione di controllo*	Measurement to monitor selected physical quantities.
Decommissioning *Ausserbetriebsetzung* *mise hors service* *dismissione*	Discontinuation or interruption of use.
Defect *Mangel* *défaut* *difetto*	Missing property, which the construction should exhibit, according to generally acknowledged rules of construction technology or according to an agreement.
Deformation capacity *Verformungsvermögen* *capacité de déformation* *capacità di deformazione*	Ability of a structure and its members to be deformed up to failure, elastically and in general also plastically.
Degree of protection *Schutzgrad* *degré de protection* *grado di protezione*	Measure for the material implementation of objectives of protection (e.g. in case of fire the assignment to fire resistance classes, in case of earthquake the assignment to construction works classes).
Design *Projektierung* *élaboration de projet* *progettazione*	Entirety of conceptual design, structural analysis and dimensioning.
Design boundary conditions *Entwurfsrandbedingungen* *conditions cadres du projet* *vincoli di progetto*	Space, time, legal, financial, constructional, material-, execution- and operation-related conditions for the conceptual design.
Detailing *Konstruktive Durchbildung* *dispositions constructives* *sviluppo dei dettagli costruttivi*	Specifying the constructional details, taking into account mutual dependencies.
Deterioration *Schädigung* *détérioration* *deterioramento*	Occurrent weakening of the material substance of a structure or other structural members taking place after acceptance.
Deterioration mechanism *Schädigungsmechanismus* *mécanisme de détérioration* *meccanismo di deterioramento*	Scientifically describable process of the cause and development of a deterioration.
Dimensioning *Bemessung* *dimensionnement* *dimensionamento*	Fixing the dimensions, the construction materials and the detailing of a structure on the basis of constructional and execution-related considerations and generally numerical verifications.
Dimensioning criteria *Bemessungskriterien* *critères de dimensionnement* *criteri di dimensionamento*	Relationships to be fulfilled between the effects of actions of relevant load cases and the corresponding ultimate resistances or serviceability limits, and between destabilising and stabilising effects of actions.
Dimensioning situations *Bemessungssituationen* *situations de dimensionnement* *situazioni di dimensionamento*	Physical situations and present conditions within a certain period of time, for which it is verified that the decisive limit states are not exceeded.

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Dimensioning value *Bemessungswert* *valeur de dimensionnement* *valore di dimensionamento*	A value, determined from a characteristic or another representative value in combination with partial or conversion factors, or specified directly, which is used in a verification.
Dismantlement *Rückbau* *déconstruction* *smantellamento*	Planned demolition or dismantling of a construction works with separation of the structural members and construction materials, fulfilling disposal requirements.
Documents for construction *Dokumente der Ausführung* *documents d'exécution* *documenti d'esecuzione*	Contract documents, construction programmes, minutes of meetings and records of construction inspections, together with the daily record of work carried out.
Ductility *Duktilität* *ductilité* *duttilità*	Plastic deformation capacity characterised by irreversible deformations and energy dissipation, usually related to the limit of the elastic behaviour.
Durability *Dauerhaftigkeit* *durabilité* *durabilità*	Fulfilment of the requirements of structural safety and serviceability within the framework of the agreed use and the service life and the foreseeable actions, without unforeseen expenditure on maintenance and repair.
Economy *Wirtschaftlichkeit* *économie* *economicità*	Moderate use of financial means and natural resources in relation to the whole period of design, execution and use.
Effects of actions *Auswirkungen* *effets des actions* *effetti delle azioni*	Response of a structure to actions (stresses, stress resultants, reactions, deformations, displacements, as well as other effects depending on the type of construction).
Environmental influences *Umwelteinflüsse* *influences de l'environnement* *influssi ambientali*	Physical, chemical or biological influences stemming from the environment, which can impair the structural safety, serviceability and durability.
Estimate *Erwartungswert* *valeur probable* *valore medio*	Generally mean value of a stochastic quantity.
Examination *Überprüfung* *examen* *esame*	Condition survey and evaluation including recommendation of remedial measures occasioned by special circumstances.
Execution *Ausführung* *exécution* *esecuzione*	All the activities and measures involved in physically creating a construction, including preparation for execution.
Failure *Versagen* *défaillance* *cedimento strutturale*	Exhaustion of the ultimate resistance due to instability, rupture or fatigue.
Fatigue resistance *Ermüdungsfestigkeit* *résistance à la fatigue* *resistenza alla fatica*	Ultimate resistance under frequently repeated actions.

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Fixed action *Ortsfeste Einwirkung* *action fixe* *azione fissa*	Action with fixed distribution over the structure or structural member.
Free action *Freie Einwirkung* *action libre* *azione libera*	An action whose distribution over the structure is not fixed.
Geometrical properties *Geometrische Grössen* *données géométriques* *grandezze geometriche*	Planned dimensions and unwanted imperfections of a structure.
Ground *Baugrund* *sol de fondation* *terreno di fondazione*	Soil or rock in the area of a planned construction works or foundation.
Hazard *Gefährdung* *danger* *rischio*	A condition which calls into question the structural safety.
Hazard scenario *Gefährdungsbild* *situation de risque* *situazione di rischio*	Critical situation characterised by a leading hazard and accompanying circumstances.
Inspection *Inspektion* *inspection* *ispezione*	Determining the condition by specific, and as a rule, visual and simple investigations and condition assessment.
Integration *Einpassung* *intégration* *integrazione*	Adaptation of a construction works to the natural and manmade environment.
Leading action *Leiteinwirkung* *action prépondérante* *azione preponderante*	Main action in a load case.
Leading hazard *Leitgefahr* *danger prépondérant* *rischio preponderante*	Main hazard in a hazard scenario.
Limit state *Grenzzustand* *état-limite* *stato limite*	State reached at which the requirements of structural safety or serviceability are just fulfilled.
Load case *Lastfall* *cas de charge* *caso di carico*	A physically compatible combination of simultaneously occurring actions considered for a specific verification.
Long-term effects *Langzeitwirkungen* *effets à long terme* *effeti di lunga durata*	Effects in the form of long-term changes of stress or strain, e.g. creeping, shrinkage, relaxation or consolidation.

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Maintenance *Instandhaltung* *entretien* *manutenzione*	Preserving the serviceability by means of simple and regular measures.
Maintenance plan *Unterhaltsplan* *plan d'entretien* *piano di manutenzione*	Instructions for the maintenance specific to the construction works.
Materials *Werkstoffe* *matériaux* *materiali*	Metal, non-metallic-inorganic or organic material with useful technical properties.
Method of construction *Bauverfahren* *procédé d'exécution* *procedimento esecutivo*	Manner in which the construction is carried out.
Modification *Veränderung* *modification* *modifica*	Making changes in a construction works for the purpose of adapting it to new requirements.
Monitoring *Überwachung* *surveillance* *monitoraggio*	Determination and assessment of the condition with recommendations on steps to be taken.
Monitoring plan *Überwachungsplan* *plan de surveillance* *piano di monitoraggio*	Instructions for the monitoring specific to the construction works.
Nominal value *Nennwert* *valeur nominale* *valore nominale*	A value specified on a non-statistical basis, e.g. based on experience or on physical conditions, or a planned prescribed value.
Objective of protection *Schutzziel* *objectif de protection* *obiettivo della protezione*	Qualitative and quantitative specification of the requirements of a structure for the case of accidental occurrences and conditions.
Observation *Beobachtung* *observation* *osservazione*	Examining the serviceability by simple and regular checks.
Observation method *Beobachtungsmethode* *méthode observationnelle* *metodo d'osservazione*	Possible procedure in the case of insufficiently reliable basic information for the design, execution and use of a structure, involving certain acceptable risks, a prediction of behaviour, and the specification of associated limit values together with corresponding monitoring and safety measures.
Operation instructions *Betriebsanweisungen* *instructions d'exploitation* *istruzioni per l'esercizio*	Instructions for the owner and for the operating staff on the handling and operation of the technical equipment.
Overall stability *Gesamtstabilität* *stabilité d'ensemble* *stabilità globale*	State of stable equilibrium for the whole structure, considered as a rigid body.

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Permanent action
Ständige Einwirkung
action permanente
azione permanente

Action almost constant or monotonically approaching a limiting value during a reference period.

Persistent dimensioning situation
Andauernde Bemessungssituation
situation de dimensionnement durable
situazione di dimensionamento persistente

Dimensioning situation which is decisive during a period of the same order of magnitude as the service life.

Planning of intervention measures
Massnahmenplanung
étude d'intervention
pianificazione d'intervento

Systematic planning of the further steps of a structural or operational nature to be taken in accordance with the basic decision of the owner.

Preparation for construction
Vorbereitung der Ausführung
préparation de l'exécution
preparazione dell'esecuzione

Invitation to tender, tendering, evaluation of tenders, conclusion of contract for materials and work, as well as preparation of execution work.

Preservation
Erhaltung
maintenance
conservazione

All the activities and measures undertaken to ensure the continued existence of a construction works, including its material and non-material values.

Protection and precaution measures
Schutz- und Vorsorgemassnahmen
mesures de protection et de prévoyance
misure protettive e precauzionali

Precautions during execution for the protection of persons and their health as well as the protection of property.

Prudent estimate
Vorsichtiger Erwartungswert
valeur probable prudente
valore prudenziale

A value which compared to the estimate is provided with an adequate margin to meet the required reliability.

Recommendation of intervention measures
Massnahmenempfehlung
recommandation d'intervention
raccomandazione d'intervento

Recommendations resulting from the condition evaluation intended to help the owner make the basic decision on further steps to be taken.

Reference period
Bezugszeitraum
durée de référence
periodo di riferimento

Period for the statistical determination of actions.

Rehabilitation
Instandsetzung
remise en état
ripristino

Restoring the structural safety and serviceability for an agreed period.

Reliability
Zuverlässigkeit
fiabilité
affidabilità

In general, a probabilistically formulated measure for fulfilling the requirements of structural safety and serviceability.

Report of intervention measures
Massnahmenbericht
rapport d'intervention
rapporto d'intervento

Report on the preservation measures carried out, including details of objectives, participants, methods, products, tests, results and costs.

Representative value
Repräsentativer Wert
valeur représentative
valore rappresentativo

The value of an action used in a verification (generally characteristic value, for variable actions also occasional, frequent or quasi-permanent value).

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Resistance *Tragfähigkeit* *résistance* *resistenza*	Ability of a structure and its members to withstand actions during execution and use.
Robustness *Robustheit* *robustesse* *robustezza*	Ability of a structure and its members to keep the amount of deterioration or failure within reasonable limits in relation to the cause.
Service criteria agreement *Nutzungsvereinbarung* *convention d'utilisation* *convenzione d'utilizzazione*	Description of the builders' or the owners' utilisation and protection aims as well as the basic conditions, requirements and regulations for the design, execution, use and preservation of the construction.
Service instructions *Nutzungsanweisungen* *instructions d'utilisation* *istruzioni per l'utilizzazione*	Instructions for the owner and the user on the use of the construction.
Service life *Nutzungsdauer* *durée d'utilisation* *durata d'utilizzazione*	Agreed period after commissioning, for which a structure or a structural member can be used for its intended purpose, subjected to monitoring and maintenance according to the monitoring plan and the maintenance plan.
Service requirements *Nutzungsanforderungen* *exigences d'utilisation* *esigenze d'utilizzazione*	Requirements of the properties and behaviour of a construction works resulting from the agreed use.
Service situations *Nutzungszustände* *états d'utilisation* *stati d'utilizzazione*	Physical circumstances and present conditions during the service life.
Serviceability *Gebrauchstauglichkeit* *aptitude au service* *efficienza funzionale*	Ability of a structure and other members to ensure functionality and appearance of the construction works as well as user comfort, in accordance with serviceability limits.
Serviceability limit *Gebrauchsgrenze* *limite de service* *limite di servizio*	Agreed limit of serviceability.
Serviceability limit state *Grenzzustand der Gebrauchstauglichkeit* *état-limite de service* *stato limite di servizio*	State when a serviceability limit is reached.
Static calculations *Statische Berechnung* *calculs statiques* *calcolo statico*	Presentation of the structural analysis and dimensioning.
Structural analysis *Tragwerksanalyse* *analyse structurale* *analisi strutturale*	Determination of effects of actions by means of a structural model, for the structure as a whole and its individual structural members.
Structural concept *Tragwerkskonzept* *concept de la structure* *concetto strutturale*	The basic idea underlying the structural design.

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Structural member *Bauteil* *élément de construction* *elemento strutturale*	Physically distinguishable part of a construction.
Structural model *Tragwerksmodell* *modèle de la structure* *modello strutturale*	Delimitation and idealization of the structural system for structural analysis.
Structural safety *Tragsicherheit* *sécurité structurale* *sicurezza strutturale*	Ability of a structure and its members to guarantee the overall stability as well as an adequate ultimate resistance (including fatigue resistance), corresponding to the assumed actions and the required reliability.
Structural system *Tragsystem* *système structural* *sistema strutturale*	Arrangement of structural members as well as the manner in which they interact.
Structure *Tragwerk* *structure porteuse* *struttura portante*	Totality of the structural members together with the ground, which are required to maintain the equilibrium and the shape of a construction works.
Supplementary safety measures *Ergänzende Sicherheitsmassnahmen* *mesures de sécurité supplémentaires* *misure di sicurezza supplementari*	Measures in order to protect persons, environment and construction works, to be adopted in case of assumption of structural safety not being in compliance with code requirements.
Technical report *Technischer Bericht* *rapport technique* *relazione tecnica*	Report about the design work.
Tender documents *Ausschreibungsunterlagen* *documents de soumission* *documenti per l'appalto*	Text of the planned contract for materials and work, special conditions, bill of quantities or work description, plans and general conditions.
Transient dimensioning situation *Vorübergehende Bemessungssituation* *situation de dimensionnement transitoire* *situazione di dimensionamento transitoria*	Dimensioning situation which is decisive during a much shorter period than the service life.
Type of construction *Bauweise* *mode de construction* *tipo di costruzione*	Class of construction characterised by the main materials used, e.g. reinforced concrete, steel, composite steel and concrete, timber, masonry.
Ultimate limit state *Grenzzustand der Tragsicherheit* *état-limite ultime* *stato limite ultimo*	State at which the limit of structural safety is reached.
Ultimate resistance *Tragwiderstand* *résistance ultime* *resistenza ultima*	Limit of resistance.
Urgent safety measures *Sichernde Sofortmassnahmen* *mesures urgentes de sécurité* *misure urgenti di sicurezza*	Measures put in place immediately for the protection of persons, the environment and the construction works as a result of monitoring or examination.

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Use *Nutzung* *utilisation* *utilizzazione*	Utilisation of the construction works as described in the service criteria agreement and in the basis of design.
Variable action *Veränderliche Einwirkung* *action variable* *azione variabile*	Action which is not permanently acting, not constant or not monotonically changing during a reference period.
Verification *Nachweis* *vérification* *verifica*	Confirmation of the fulfilment of a dimensioning criterion.

1.2 Symbols

1.2.1 Latin upper-case letters (partly in combination with reduction factors ψ)

A_d	dimensioning value of an accidental action
C_d	dimensioning value of a serviceability limit
E_d	dimensioning value of an action effect
$E_{d,dst}$	dimensioning value of a destabilising action effect
$E_{d,stb}$	dimensioning value of a stabilising action effect
$E\{\ldots\}$	action effect as function of the dimensioning values within the brackets [1)]
F_d	dimensioning value of an action
F_k	characteristic value of an action
F_{rep}	representative value of an action
G	permanent action
G_k	characteristic value of a permanent action
$G_{k,inf}$	lower characteristic value of a permanent action
$G_{k,sup}$	upper characteristic value of a permanent action
H	height of building
P_k	characteristic value of an action due to prestressing
Q_k	characteristic value of a variable action
Q_{k1}	characteristic value of the (variable) leading action
$\psi_0 Q_k$	occasional value of a variable action
$\psi_{0i} Q_{ki}$	occasional value of the variable accompanying action i
$\psi_1 Q_k$	frequent value of a variable action
$\psi_{11} Q_{k1}$	frequent value of a variable action in combination with an accidental action or frequent value of the (variable) leading action
$\psi_2 Q_k$	quasi-permanent value of a variable action

[1)] Depending on the verification some of these dimensioning values may not apply.

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$\psi_{2i}Q_{ki}$ quasi-permanent value of the variable action *i* in combination with an accidental action or the frequent value of the (variable) leading action

R_d dimensioning value of the ultimate resistance

R_k characteristic value of the ultimate resistance

$R\{\ldots\}$ ultimate resistance as function of the dimensioning values within the brackets [1)]

X_d dimensioning value of a construction material or ground property

X_k characteristic value of a construction material or ground property

1.2.2 Latin lower-case letters

a_d dimensioning value of a geometrical property

a_{nom} nominal value of a geometrical property

f eigenfrequency

h storey height

h_0 reference height according to code SIA 261

l span or double cantilever length

u horizontal displacement

v design speed

w deflection

1.2.3 Greek upper-case letters

Δa difference of a geometrical property

Δu difference of displacement

Φ dynamic factor

1.2.4 Greek lower-case letters

α factor to calibrate standard load models

α_t track twist

γ_f partial factor for actions (takes into account possible unfavourable deviations from representative value)

γ_F load factor (= $\gamma_S\,\gamma_f$)

γ_G load factor for a permanent action

$\gamma_{G,inf}$ load factor for permanent actions (lower value)

$\gamma_{G,sup}$ load factor for permanent actions (upper value)

γ_m partial factor for a construction material or ground property (takes into account possible unfavourable deviations from the characteristic value)

γ_M resistance factor (= $\gamma_R\,\gamma_m$)

γ_P load factor for an action due to prestressing

γ_Q load factor for a variable action

γ_{Q1} load factor for the (variable) leading action

$\gamma_{Q,inf}$ load factor for a variable action (earth or water pressure, lower value)

[1)] Depending on the verification some of these dimensioning values may not apply.

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$\gamma_{Q,sup}$ load factor for a variable action (earth or water pressure, upper value)

γ_R partial factor for the ultimate resistance (takes into account uncertainties in the resistance model)

γ_S partial factor for effects of actions (takes into account uncertainties in the model with respect to actions and effects of actions)

δ_v vertical relative displacement at expansion joints or of deck ends behind abutments

η conversion factor

ψ reduction factor

ψ_0 reduction factor for the occasional value of a variable action

ψ_1 reduction factor for the frequent value of a variable action

ψ_2 reduction factor for the quasi-permanent value of a variable action

1.3 Units

1.3.1 SI units shall be used in design.

1.3.2 The following units are recommended:

length, displacement	m, mm
area	m^2, mm^2
volume, moment of resistance, static moment	m^3, mm^3
moment of inertia (second moment of area)	m^4, mm^4
angle	° (degree), rad
curvature	1/m, mrad/m
time	s
duration, age	min, h, d, year
velocity (speed), coefficient of permeability	m/s
coefficient of consolidation	m^2/s
acceleration	m/s^2
frequency	Hz
mass	kg, t (tonne)
density	kg/m^3
force, load	kN, MN
moment	kNm, MNm
stress, strength, adhesion, cohesion	N/mm^2, kN/m^2
Young's modulus, shear modulus	kN/mm^2, MN/m^2
line load	kN/m
surface load	kN/m^2
unit weight, body force	kN/m^3, MN/m^3
pressure	kPa, MPa
work	J
temperature	°C, K

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2 CONCEPTUAL DESIGN

2.1 General

2.1.1 Starting from the service criteria the conceptual design should lead to a suitable structural concept.

2.1.2 The service criteria shall be formulated in the service criteria agreement.

2.1.3 The bases and requirements for the subsequent design, execution, use and preservation that derive from the conceptual design shall be described in the basis of design.

2.2 Service criteria agreement

2.2.1 The service criteria agreement shall be formulated on the basis of a dialogue between the owner and the design engineers.

2.2.2 The service criteria agreement comprises:
- general aims for the use (service situations and service life) of the construction works
- ambient conditions and third-party requirements
- operational and maintenance requirements
- special requirements of the owner
- objectives of protection and special risks
- code-related requirements.

2.2.3 The objectives and the degree of protection shall be determined on the basis of a risk evaluation.

2.3 Requirements

2.3.1 Apart from being appropriately integrated and configured, a structure should be economic, robust, reliable and durable during its service life.

2.3.2 The service life shall be agreed. Possible standard values are:

– temporary structures	up to 10 years
– replaceable members	up to 25 years
– buildings and other construction works of average importance	50 years
– construction works of greater importance	100 years.

2.3.3 Fixing the required reliability is a matter of experience augmented by theoretical considerations. Normally one proceeds from the verification concept specified in Section 4.4.

2.3.4 The required reliability may be differentiated depending on the limit state and reference period, either for the whole structure or for its individual structural members.

2.3.5 Fixing specific levels of reliability may depend on:
- the type and consequences of failure
- the amount of acceptable damage
- the importance of the construction in dealing with a catastrophe after an accidental event
- the expenditure to reduce the risk
- the possibilities of monitoring, maintenance and repair as well as the corresponding expenditure.

2.3.6 Ensuring an adequate reliability requires in particular:
- consideration of uncertainties regarding actions, structural modelling and the determination of effects of actions
- quality assurance measures during design, execution, use and preservation.

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2.3.7 Ensuring the durability during the service life requires in particular:
- identifying the relevant actions
- estimating possible deterioration of the structure
- adequate measures during design, execution, use and preservation, especially constructional and construction material-technological measures to protect the construction materials and structural members, an expert execution as well as planned monitoring and maintenance.

2.4 Procedure

2.4.1 The conceptual design involves preparing different alternatives, taking account of the relevant boundary conditions, examining them for their feasibility and assessing the alternatives with respect to fulfilling the requirements.

2.4.2 The design boundary conditions shall be clarified and their relevance examined. They include for example:
- location, hazard zone, topography, alignment, clearances, distances to boundary, minimum and maximum dimensions
- restrictions regarding the design and construction times and the service life
- legal aspects (laws, ordinances, directives)
- budget limitations
- quality, availability and reusability of construction materials
- ground properties
- applicability of construction methods, transport and assembly possibilities
- maintaining the use of traffic structures and lifelines (pipelines, etc.)
- equipment and facilities for monitoring and maintenance measures.

2.4.3 In the conceptual design, the decisive actions and effects of actions as well as the relevant hazards and means of controlling them shall be identified.

2.4.4 Possible hazard scenarios shall be thought through and suitable measures shall be specified in order to keep the hazards under control or to limit them to an acceptable amount.

2.4.5 The following influences, for example, may represent a hazard:
- deviation from the assumed values of the actions
- actions imposed by the ground
- actions due to gravitational or meteorological natural hazards
- physical actions, e.g. due to humidity or temperature
- chemical actions, e.g. due to de-icing salt or due to ground water
- biological action, e.g. due to insects or mushrooms
- resonance effects
- deviations from the planned values of the ultimate resistances of the structure or the ground
- impairment of the ultimate resistance through corrosion, increasing brittleness or fatigue
- impairment of the ultimate resistance due to fire, explosion, impact, rupture of pipes, or earthquakes.

2.4.6 Hazards can be mitigated by means of one or more of the following measures:
- elimination, prevention or hazard reduction
- controls or alarm systems
- choice of structural systems which are less susceptible to the hazards under consideration
- choice of structural systems which can tolerate local damage as well as the loss of individual structural members without global failure
- choice of structural systems which do not fail without prior warning
- limitation of the spread of fire to fire sections
- choice of suitable construction materials
- appropriate structural analysis and dimensioning
- careful detailing
- execution carried out as planned and with the necessary care
- planning special protection measures
- appropriate monitoring and maintenance.

2.4.7 The foreseeable service situations shall be thought through and appropriate measures to ensure serviceability shall be specified.

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2.4.8 In the conceptual design, experience gained with comparable construction works shall be included.

2.4.9 In evaluating the alternatives, special attention shall be given to simplicity of execution, insensitivity to unavoidable inaccuracies and possible mistakes in execution as well as adaptability to possible changes in execution and use.

2.5 Structural concept and basis of design

2.5.1 The structural concept includes:
- the chosen structural system
- information on the most important dimensions, construction material properties and constructional details
- comments on the envisaged methods of construction.

2.5.2 The basis of design describes:
- the hazard scenarios considered
- the requirements of structural safety, serviceability and durability together with the measures needed to guarantee them, including division of responsibilities, processes, controls and corrective mechanisms
- the assumed ground conditions
- the important assumptions in the structural and analytical models
- the accepted risks
- other conditions relevant to the design.

2.5.3 The extent and content of the basis of design shall be adapted to the importance of the construction works and the associated hazards and environmental risks.

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3 STRUCTURAL ANALYSIS

3.1 General

3.1.1 The structural analysis should consider the behaviour of a structure in relation to the dimensioning situations, taking into account the relevant factors.

3.1.2 The methods of structural analysis shall be based on established, if necessary experimentally-confirmed, theory and engineering practice.

3.1.3 The basis and the results of the structural analysis shall be set out in the static calculations and in the technical report.

3.1.4 The results of the structural analysis shall be checked for plausibility.

3.2 Actions

3.2.1 Classification

3.2.1.1 Actions shall be differentiated according to their origin (resulting from execution, use or environmental influences) and their type (mechanical, other physical, chemical or biological actions).

3.2.1.2 Actions shall also be classified according to:
- time-dependence (permanent, variable or accidental actions)
- place-dependence (fixed or free actions)
- dynamic effect (static or dynamic actions).

3.2.1.3 Forces or constrained deformations that are applied in a controlled way for the purpose of prestressing, shall be treated as permanent actions.

3.2.1.4 Free actions shall be specified in regard to their position, magnitude and direction.

3.2.1.5 For fatigue verifications and dynamic calculations information on the time variation of individual relevant actions is necessary.

3.2.2 Characteristic values

3.2.2.1 An action is described by a model and its magnitude is fixed by one or more scalar quantities, which can assume different representative values F_{rep}.

3.2.2.2 The characteristic value F_k of an action is the most important representative value and shall be specified as follows:
- according to SIA 261 and 267 as mean value, as upper and lower value, as nominal value, or
- design-specific, taking into consideration the requirements of SIA 261.

3.2.2.3 For permanent actions with a small range of variation (coefficient of variation ≤ 5%) a single value G_k (usually the mean value) may be used, if the structure does not react sensitively to changes in G. For sensitive structures or greater range of variation, lower and upper values $G_{k,inf}$ and $G_{k,sup}$ shall be used (in general the 5% and 95% fractile values of the statistical distribution of G, which may be assumed to exhibit a Gaussian normal distribution).

3.2.2.4 Dead loads of structures may in general be represented by a single characteristic value and may be determined from the planned dimensions as well as the mean unit weights (see SIA 261).

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3.2.2.5 As characteristic values of actions due to prestressing, P_k, depending on the considered point in time and the type of prestressing, one shall use mean values, upper or lower values.

3.2.2.6 In the case of variable actions the characteristic value Q_k shall be specified as follows:
– as upper or lower value, which may not be exceeded or reached with a specified probability during the reference period
– as nominal value, if the statistical distribution is unknown.

Usually the reference period may be assumed to be one year and the specified probability to be 98% (corresponding to a return period of 50 years).

3.2.2.7 Actions due to water shall be determined based on measurements or estimates of water levels or pore water pressures and their variation, taking into account the hydraulic potential. The decisive water levels, water pressures and pore water pressures shall be fixed according to the dimensioning situation and the reference period. Water flow and waves shall be considered, if necessary.

3.2.2.8 In the case of accidental actions the dimensioning value A_d shall be specified directly.

3.2.3 Further representative values for variable actions

3.2.3.1 The reduced probability of the simultaneous occurrence of the unfavourable values of several independent actions shall be taken into account by means of the reduction factors ψ_0, ψ_1 and ψ_2 (see Appendix A to F).

3.2.3.2 Occasional values $\psi_0 Q_k$ are used together with a variable leading action.

3.2.3.3 Frequent values $\psi_1 Q_k$ are fixed such that
– either the total time within a reference period, during which they are exceeded, is limited by a given value (e.g. 1% for live loads in buildings), or
– the frequency of exceeding is limited by a given value (e.g. once per week for road traffic loads).

3.2.3.4 Quasi-permanent values $\psi_2 Q_k$ are fixed
– either such that they are exceeded during a considerable part of the reference period (e.g. 50% for live loads in buildings), or
– as mean values for the reference period.

3.2.4 Fatigue actions

3.2.4.1 In general the provisions of SIA 261 and SIA 261/1 apply.

3.2.4.2 For cases outside the range of application of SIA 261 and SIA 261/1 fatigue actions shall be determined from measurements or equivalent investigations of action spectra.

3.2.5 Dynamic actions

3.2.5.1 The action models given in SIA 261 and SIA 261/1 consider acceleration effects either in the characteristic values or by means of dynamic factors.

3.2.5.2 If dynamic actions cause considerable structural accelerations, dynamic analyses shall be carried out.

3.2.6 Actions imposed by the ground

3.2.6.1 Actions imposed by the ground include earth and water pressures, and they are leading to effects of actions like effective normal stresses, shear forces and ground deformations. Usually they shall be treated as permanent actions. Depending on the dimensioning situation it may be appropriate to treat them as either variable or accidental actions.

3.2.6.2 Actions due to groundwater can be determined either by uplift and total water pressure (total permanent actions and absolute porewater pressure) or by seepage pressure (effective permanent actions and excess porewater pressure).

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3.2.6.3 Actions imposed by the ground shall be determined on the basis of characteristic values according to established geotechnical methods.

3.2.6.4 The magnitude, direction and distribution of earth pressures depend on the ground properties, the construction processes, the effect of the groundwater and the interaction between the ground and the remaining structure.

3.2.6.5 As a rule, by investigating the interaction between the ground and the remaining structure, prudent estimates of the stiffnesses of the ground and of the remaining structure shall be assumed. The non-linear behaviour of the ground shall be taken into account. Depending on the dimensioning situation, upper or lower values of stiffness may be decisive.

3.2.6.6 Actions due to groundwater can correspond to persistent, transient and accidental dimensioning situations and shall be determined by taking into account either the total water pressure or the corresponding water levels and seepage pressures.

3.2.7 Environmental influences

3.2.7.1 Environmental influences shall be considered in specifying the structural concept and the dimensions of the structure, in selecting the construction materials, and in detailing.

3.2.7.2 The effects of environmental influences are strongly dependent on the type of construction. As far as possible they shall be determined quantitatively according to SIA 262 to 267.

3.2.7.3 The combined action of two or more environmental influences often leads to more serious effects of actions than the sum of the effects of actions due to the individual influences.

3.3 Structural model

3.3.1 General

3.3.1.1 The structural model relates actions and geometrical, construction material and ground properties in order to carry out the structural analysis.

3.3.1.2 The chosen structural model has to be suitable for predicting the structural behaviour in the dimensioning situations considered.

3.3.2 Geometrical properties

3.3.2.1 Geometrical properties shall be represented either by characteristic values or directly by dimensioning values (e.g. for imperfections).

3.3.2.2 In the case of specified geometrical properties (dimensions) the characteristic value normally corresponds to the nominal value, for other geometrical properties (e.g. heights of fill, water levels) to the prudent estimate. If given dimensional tolerances can be exceeded, this has to be stated in the basis of design.

3.3.2.3 If the statistical distribution of geometrical properties is sufficiently well known, the dimensioning values may correspond to some given fractile values.

3.3.2.4 The imperfections and dimensional tolerances to be taken into account in the design are given in SIA 262 to 267.

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3.3.3 Construction material and ground properties

3.3.3.1 The properties of construction materials and products as well as the ground properties are represented by characteristic values, which in a hypothetically unlimited series of tests are not reached or not exceeded with a certain probability.

3.3.3.2 If the verification of a limit state is sensitive to changes in a construction material or ground property, lower or upper characteristic values (in general the 5% and 95% fractile values) shall be used.

3.3.3.3 If the verification of a limit state is relatively insensitive to changes in a construction material or ground property and this property exhibits a small scatter, the mean value may be assumed as characteristic value.

3.3.3.4 As a rule values of construction material and ground properties shall be determined by means of standard testing methods under prescribed conditions. Sometimes a conversion factor has to be used to convert the test results to values for which one can assume that they are representative for the structure or the ground.

3.3.3.5 In the case of ground properties the characteristic value usually corresponds to a prudent estimate (5% or 95% fractile value of the mean value distribution).

3.3.3.6 If insufficient statistical data is available, nominal values may be used as characteristic values, or dimensioning values can be specified directly.

3.3.3.7 In some cases upper characteristic values of the strength of construction materials or of the ground shall be used, e.g. in capacity design and for the tensile strength of concrete, when considering constrained or restrained deformations.

3.3.3.8 The fatigue resistance shall be taken into account according to SIA 262 to 265.

3.3.4 Analytical models for static actions

3.3.4.1 The chosen analytical model shall be based on suitable relationships between the governing force and deformation quantities of the individual structural members and their connections.

3.3.4.2 Second order effects shall be considered if they lead to a substantial increase of the effects of actions. Details are given in SIA 262 to 266.

3.3.4.3 For plastic deformation analyses the influence of constrained or restrained deformations shall be considered.

3.3.4.4 Constrained deformations imposed by the ground shall be taken into account using prudent estimates for the stiffness of the ground.

3.3.4.5 If the structural behaviour is considerably influenced by the ground, the interaction between ground and remaining structure shall be suitably modelled. Details are given in SIA 267.

3.3.5 Analytical models for dynamic actions

3.3.5.1 In the analytical model the mass, stiffness, ultimate resistance and damping of all structural members shall be treated as realistically as possible.

3.3.5.2 If dynamic actions can be treated quasi-statically, in the case of accidental actions they can be replaced by equivalent static forces taking into account the ductility of the structure. In the case of variable actions, the dynamic components are either added to the static values or taken into account by means of dynamic amplification factors. For certain dynamic amplification factors the eigenfrequencies have to be determined.

3.3.5.3 In the case of substantial interaction between ground and remaining structure, the contribution of the ground can be taken into account by means of suitable spring, mass and damping elements. Details are given in SIA 267.

3.3.5.4 In certain cases the actions can be determined by means of a vibration analysis assuming linear behaviour with regard to geometry and construction materials. If only the fundamental frequency is important, the vibration analysis can be replaced by a calculation with equivalent static actions, which among other factors depend on the vibration mode shape, the eigenfrequency and the damping.

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3.3.5.5 In certain cases the dynamic actions can be represented by time histories or investigated in the frequency domain.

3.3.5.6 Standard values to assess vibrations in the serviceability limit state are given in the Appendixes A and C.

3.3.6 Analytical models for fire actions

3.3.6.1 The present code and the codes SIA 261 to 266 are restricted to information concerning structural fire protection. The objective is to limit the spread of fire to the affected fire section and to provide protection for persons from a premature failure of the structure.

3.3.6.2 The fulfilment of the requirements for structural fire protection shall be verified by means of an analysis of the whole structure, individual parts of the structure, or individual structural members.

3.3.6.3 The structural analysis for the fire dimensioning situation has to be carried out using suitable models, whereby thermal and mechanical actions as well as structural behaviour at high temperatures shall be considered. The structural analysis may be supplemented by tests.

3.3.6.4 Thermal and mechanical actions shall be specified according to code SIA 261.

3.3.6.5 Structural behaviour at high temperatures shall be investigated according to the information given in codes SIA 262 to 266, or with the aid of tests.

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4 DIMENSIONING

4.1 General

4.1.1 The dimensioning comprises the specification of the dimensions, the construction materials and their properties and the detailing of a structure in accordance with the design requirements.

4.1.2 As a rule dimensioning is based on limit states. This requires:
- the development of suitable structural and action models
- the choice of appropriate analytical models
- the choice of decisive dimensioning situations
- the verification that the relevant limit states are not exceeded.

4.1.3 In general, the characteristic or other representative values defined in Section 3.2 shall be used in conjunction with the verification concept using dimensioning values according to Section 4.4.

4.1.4 If necessary, dimensioning values can be determined directly, whereby the level of reliability has to correspond at least to that underlying the verification concept specified in Section 4.4.

4.1.5 The basis and the results of the dimensioning shall be set out in the static calculations and in the technical report.

4.1.6 The results of the dimensioning shall be checked for plausibility.

4.2 Dimensioning situations

4.2.1 The dimensioning situations considered shall include all foreseeable conditions that can occur during execution and use.

4.2.2 Dimensioning situations are classified as persistent, transient and accidental situations.

4.2.3 Construction states can be considered as persistent or transient dimensioning situations.

4.2.4 Accidental dimensioning situations involve either the accidental situation itself, or they refer to the situation immediately after the accidental event.

4.2.5 The load cases belonging to a dimensioning situation shall be specified.

4.2.6 Each load case is characterised by a leading action and the simultaneous accompanying actions. As a rule, it is sufficient to consider one single variable accompanying action.

4.2.7 In case of failure without advance notice, it shall be examined, whether the effects of actions due to constrained or imposed deformation have to be considered as an additional accompanying action by using its occasional value.

4.3 Limit states

4.3.1 Ultimate limit states

4.3.1.1 Ultimate limit states concern:
- the safety of the structure
- the safety of persons.

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4.3.1.2 Ultimate limit states involve:
- the overall stability of the structure
- the ultimate resistance of the structure or one of its structural members, including supports and its foundation
- the fatigue resistance of the structure or one of its structural members.

4.3.2 Serviceability limit states

4.3.2.1 Serviceability limit states concern:
- the functionality of the construction works
- the comfort of persons using the construction works
- the appearance of the construction works.

4.3.2.2 Serviceability dimensioning criteria may be applied to:
- deformations that impair the functionality or appearance of the construction works or which cause deterioration to the structural members
- vibrations that restrict the functionality of the construction works or impair user comfort
- defective sealing that restricts the functionality or impairs the use of the construction works
- construction-specific effects of actions (e.g. cracks, slip in connections) that impair the appearance of the construction works and the durability of the structure
- limit values of environmental pollution (e.g. rise of groundwater, flooding).

4.4 Verifications

4.4.1 General

4.4.1.1 Basically the structural safety and the serviceability have to be verified. The verifications shall be carried out with the aid of dimensioning values.

4.4.1.2 A verification can be dispensed with, if it is clear that it is not decisive.

4.4.1.3 A verification can be dispensed with, if it can be shown that the corresponding requirements are of secondary importance or can be achieved by constructional or execution measures. The chosen procedure shall be documented in the basis of design.

4.4.2 Dimensioning values

4.4.2.1 The dimensioning value F_d of an action is together with (9):

$$F_d = \gamma_f F_{rep} \quad (1)$$

or together with (10):

$$F_d = \gamma_F F_{rep} \quad (2)$$

with the load factor:

$$\gamma_F = \gamma_S \gamma_f \quad (3)$$

The partial factor γ_S takes into account the model uncertainties with respect to actions and effects of actions.

The partial factor γ_f takes into account possible unfavourable deviations of the action from the representative value F_{rep}.

4.4.2.2 The dimensioning value A_d for earthquake action is specified according to SIA 261, taking into consideration the structural behaviour and other criteria.

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4.4.2.3 Usually, for the dimensioning value of a geometrical property a_d the following applies:

$$a_d = a_{nom} \tag{4}$$

If deviations from the nominal value exert a considerable influence on the reliability of the structure, this shall be taken into account as follows:

$$a_d = a_{nom} \pm \Delta a \tag{5}$$

4.4.2.4 The dimensioning value X_d of a construction material or ground property is together with (11):

$$X_d = \frac{\eta X_k}{\gamma_m} \tag{6}$$

or together with (12) and (13):

$$X_d = \frac{\eta X_k}{\gamma_M} \tag{7}$$

with the resistance factor:

$$\gamma_M = \gamma_R \, \gamma_m \tag{8}$$

The partial factor γ_R takes into account the uncertainties in the resistance model. The partial factor γ_m takes into account unfavourable deviations of a construction material or ground property from its respective characteristic value X_k. The conversion factor η, for example, takes into account humidity or temperature influences, scale effects, the type of failure or the duration of an action.

4.4.2.5 The dimensioning value E_d of an action effect is generally determined as:

$$E_d = \gamma_S E\{\gamma_f F_{rep}, X_d, a_d\} \tag{9}$$

Usually, instead of (9) the equation:

$$E_d = E\{\gamma_F F_{rep}, X_d, a_d\} \tag{10}$$

can be used.

4.4.2.6 If one has to distinguish between favourable and unfavourable effects of a permanent action, two load factors $\gamma_{G,inf}$ and $\gamma_{G,sup}$ shall be used. If earth and water pressures are regarded as variable actions (see section 3.2.6.1), two load factors $\gamma_{Q,inf}$ and $\gamma_{Q,sup}$ shall be used.

4.4.2.7 The dimensioning value R_d of the ultimate resistance is generally determined as:

$$R_d = \frac{R\{X_d, a_d\}}{\gamma_R} \tag{11}$$

Usually, instead of (11) the expressions:

$$R_d = R\left\{\frac{\eta X_k}{\gamma_M}, a_d\right\} \tag{12}$$

or:

$$R_d = \frac{\eta R_k}{\gamma_M} \tag{13}$$

can be used.

4.4.2.8 The dimensioning values C_d of serviceability limits shall, as a rule, be specified or agreed upon in accordance with the service criteria specific to a project. Standard values are given in the Appendices A to F. Further standard values, relating to the type of construction, are given in codes SIA 262 to 267.

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4.4.3 **Verification of structural safety**

4.4.3.1 In the verification of structural safety four types of limit state shall be distinguished:
- Type 1 involves the loss of overall stability of a structure (sliding, overturning, lift-off, or floating as a rigid body)
- Type 2 involves reaching of the ultimate resistance of the structure or one of its structural members (failure due to rupture, excessive deformations, the structure forming a mechanism, or loss of stability of one or multiple structural members)
- Type 3 involves reaching of the ultimate resistance of the ground (slides, slope failure, collapse of ground surface)
- Type 4 involves reaching the fatigue resistance of the structure or one of its structural members.

4.4.3.2 For limit states of Type 1 the structural safety can be considered as verified, if the following dimensioning criterion is fulfilled:

$$E_{d,dst} \leq E_{d,stb} \tag{14}$$

4.4.3.3 For limit states of Types 2 and 3 the structural safety can be considered as verified, if the following dimensioning criterion is fulfilled:

$$E_d \leq R_d \tag{15}$$

Table 1: Load factors for the verification of structural safety

Actions	γ_F	Limit state		
		Type 1	Type 2	Type 3
Permanent actions				
– acting unfavourably	$\gamma_{G,sup}$	1,10 [1)]	1,35 [1)]	1,00
– acting favourably	$\gamma_{G,inf}$	0,90 [1)]	0,80 [1)]	1,00
Variable actions				
– in general	γ_Q	1,50	1,50	1,30
– road traffic loads	γ_Q	1,50	1,50	1,30
– rail traffic loads:				
– load models 1, 2, 4 bis 7	γ_Q	1,45	1,45	1,25
– load model 3	γ_Q	1,45	1,20	1,25
Actions imposed by the ground				
Soil surcharge				
– acting unfavourably	$\gamma_{G,sup}$	1,10	1,35 [2) 3)]	1,00
– acting favourably	$\gamma_{G,inf}$	0,90	0,80	1,00
Earth pressure				
– acting unfavourably	$\gamma_{G,Q,sup}$	1,35	1,35	1,00
– acting favourably [4)]	$\gamma_{G,Q,inf}$	0,80	0,70	1,00
Water pressure				
– acting unfavourably	$\gamma_{G,Q,sup}$	1,05	1,20 [3)]	1,00
– acting favourably	$\gamma_{G,Q,inf}$	0,95	0,90	1,00

[1)] G is either multiplied by $\gamma_{G,sup}$ or by $\gamma_{G,inf}$, depending on whether the overall action effect is unfavourable or favourable.
[2)] For heights of fill from 2 to 6 m, $\gamma_{G,sup}$ may be reduced linearly from 1,35 to 1,20.
[3)] When using the observational method, according to code SIA 267, reduced values are admissible in certain cases.
[4)] For passive earth pressure acting favourably, $F_d = R_d$ according to code SIA 267.

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4.4.3.4 For persistent and transient dimensioning situations the dimensioning values of effects of actions according to (10) shall be determined as follows:

$$E_d = E\{\gamma_G G_k,\ \gamma_P P_k,\ \gamma_{Q1} Q_{k1},\ \psi_{0i} Q_{ki},\ X_d,\ a_d\} \quad (16)$$

The load factors γ_G and γ_Q and the reduction factors ψ_{0i} are given in Table 1 and in the Appendices A to F, respectively. The load factors γ_P are given in codes SIA 262 and 267.

For construction states an appropriate reduction of the load factors is admissible when using the observational method.

4.4.3.5 For accidental dimensioning situations the dimensioning values of effects of actions according to (10) shall in general be determined as follows:

$$E_d = E\{G_k,\ P_k,\ A_d,\ \psi_{2i} Q_{ki},\ X_d,\ a_d\} \quad (17)$$

The reduction factors ψ_{2i} are given in the Appendices A to F.

4.4.3.6 For the impact dimensioning situation, it must be decided from case to case, whether, in addition to the accidental leading action A_d and the permanent and quasi-permanent accompanying actions, a variable accompanying action with its frequent value $\psi_{11} Q_{k1}$ has to be considered.

4.4.3.7 For the fire dimensioning situation, the accidental leading action A_d considers the temperature influence on the construction material properties and the effects of the deformational behaviour under increased temperatures caused by fire.

4.4.3.8 For limit states of Type 4 the structural safety can be considered as verified, if the following dimensioning criterion is fulfilled:

$$E_d \le R_d \quad (18)$$

Codes SIA 262 to 265 and SIA 267 together with code SIA 261 govern the determination of the fatigue effects under operating loads and specify the associated resistances and partial factors.

4.4.4 Verification of serviceability

4.4.4.1 Verifications of serviceability shall be carried out for persistent and transient dimensioning situations, and in exceptional circumstances also for accidental dimensioning situations (e.g. earthquake for Construction Works Class III according to code SIA 261).

4.4.4.2 The serviceability can be considered as verified, if the following dimensioning criterion is fulfilled:

$$E_d \le C_d \quad (19)$$

Here E_d designates the decisive action effect due to the load cases considered in the investigated dimensioning situation, and C_d is the corresponding serviceability limit.

4.4.4.3 In general, E_d shall be determined according to (10), setting $\gamma_F = \gamma_m = 1{,}0$.

4.4.4.4 For persistent and transient dimensioning situations three types of load cases shall be distinguished and the corresponding decisive effects of actions shall be determined as follows:

– occasional load cases

$$E_d = E\{G_k,\ P_k,\ Q_{k1},\ \psi_{0i} Q_{ki},\ X_d,\ a_d\} \quad (20)$$

– frequent load cases

$$E_d = E\{G_k,\ P_k,\ \psi_{11} Q_{k1},\ \psi_{2i} Q_{ki},\ X_d,\ a_d\} \quad (21)$$

– quasi-permanent load cases

$$E_d = E\{G_k,\ P_k,\ \psi_{2i} Q_{ki},\ X_d,\ a_d\} \quad (22)$$

The reduction factors ψ are given in the Appendices A to F.

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4.4.4.5 For the earthquake dimensioning situation the following applies

$$E_d = E\{G_k, P_k, A_d, \psi_{2i} Q_{ki}, X_d, a_d\} \quad (23)$$

For A_d a value reduced by 50% shall be employed compared to that for the verification of structural safety, in order to take into account the reduced recurrence period for the verification of serviceability. The reduction factors ψ_{2i} are given in the Appendices A to F. For buildings with fittings exhibiting brittle and ductile behaviour the horizontal displacements per floor shall be limited to $^1/_{500}$ and $^1/_{200}$ of the floor height, respectively.

4.5 Dimensioning assisted by testing

4.5.1 The dimensioning may be supported by testing, e.g. if:
- actions, construction material or ground properties are insufficiently known
- no appropriate analytical models are available
- a large number of the same structural members are used.

4.5.2 The tests shall be planned, executed and evaluated such that the structure exhibits the same level of reliability with respect to all decisive dimensioning situations and limit states, as is usually achieved on the basis of a dimensioning according to codes SIA 260 to 267.

4.5.3 Sampling and testing conditions shall be representative.

4.5.4 Implicit safety specifications (e.g. neglecting the tensile strength of concrete) in comparable provisions of codes SIA 261 to 267 shall be considered in the evaluation of the test results.

4.6 Reliability theory

4.6.1 The dimensioning may be supported by applying methods of reliability theory, e.g. to ensure an appropriate robustness.

4.6.2 Methods of reliability theory shall be applied when leaving the range of application of SIA 261 to 267, e.g.
- for actions and hazards lying outside the range covered by code SIA 261
- for extraordinary structural systems or an extraordinary geometry of a structure
- when using construction materials and combinations of construction materials outside the usual range of experience
- in cases where failure would lead to very serious consequences.

4.6.3 Methods of reliability theory can be employed both for the dimensioning of structures or individual structural members and for the development of project-specific design specifications.

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APPENDIX A (normative)
BUILDINGS

Table 2: Reduction factors for buildings

Actions	ψ_0	ψ_1	ψ_2
Live loads in buildings			
– Category A domestic, residential areas	0,7	0,5	0,3
– Category B office areas	0,7	0,5	0,3
– Category C congregation areas	0,7	0,7	0,6
– Category D shopping areas	0,7	0,7	0,6
– Category E storage areas	1,0	0,9	0,8
Traffic loads in buildings			
– Category F vehicles under 3,5 t	0,7	0,7	0,6
– Category G vehicles from 3,5 to 16 t	0,7	0,5	0,3
– Category H roofs	0	0	0
Snow loads	$1 - 60/h_0$	$1 - 250/h_0$	$1 - 1000/h_0$
Wind forces	0,6	0,5	0
Temperature actions	0,6	0,5	0
Actions imposed by the ground			
– earth pressure	0,7	0,7	0,7
– water pressure	0,7	0,7	0,7

Table 3: Standard values for deflections of floors and beams

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21)	quasi-permanent (22)
Functionality			
– fittings with brittle behaviour	$w \le l/500$ [1) 2)]		
– fittings with ductile behaviour		$w \le l/350$ [1)]	
– use and operation		$w \le l/350$ [3)]	
Comfort		$w \le l/350$ [3)]	
Appearance			$w \le l/300$ [4)]

[1)] Deflection due to the actions and long-term effects after installation of all secondary structural members or technical equipment.

[2)] If fittings or equipment are especially sensitive to structural deformations, besides or instead of dimensioning measures, above all constructional measures shall be implemented to prevent deterioration.

[3)] Deflection due to variable actions.

[4)] Deflection after deduction of hogging, if present. Any long-term effects shall be considered.

The deflections shall be determined according to codes SIA 262 to 266.

Deviating limit values for deflections can be agreed upon in accordance with the service criteria and must be specified in the basis of design. In particular, for secondary structural members less stringent requirements can apply.

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Table 4: Standard values for horizontal displacements of walls, frames and columns

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21)	quasi-permanent (22)
Functionality – fittings with brittle behaviour – fittings with ductile behaviour – use and operation	$u \leq h/500$ [1) 2)]	 $u \leq h/200$ [1)] $u \leq H/300$	
Appearance			$u \leq h/250$ [1)]
[1)] Horizontal displacement due to variable actions. [2)] If fittings or equipment are especially sensitive to structural deformations, besides or instead of dimensioning measures above all constructional measures shall be implemented to prevent deterioration.			
The horizontal displacements shall be determined according to codes SIA 262 to 266. Deviating limit values for horizontal displacements can be agreed upon in accordance with the service requirements and must be specified in the basis of design. In particular, more stringent requirements can apply for crane supporting structures, see Appendix F.			

Table 5: Standard values for eigenfrequencies for buildings

Limit state	Eigenfrequency [Hz]
Comfort – gymnasiums and sports halls – dance and concert halls	 $f > 8{,}0$ $f > 7{,}0$
Eigenfrequencies deviating from the given standard values are admissible, provided a more exact dynamic analysis is carried out taking damping into consideration.	

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APPENDIX B (normative)
ROAD BRIDGES

Table 6: Reduction factors for road bridges

Actions	ψ_0	ψ_1	ψ_2
Vertical loads – Load Model 1 – Load Model 3 Horizontal forces [1)]	 0,75 0 0,75	 0,75 0 0,75	 0 0 0
Snow loads [2)]	0,6	0,2	0
Wind forces	0,6	0,2	0
Temperature actions	0,6	0,6	0,5
Actions imposed by the ground – earth pressure – water pressure	 0,7 0,7	 0,7 0,7	 0,7 0,7

[1)] The horizontal forces shall be assumed to act simultaneously with Load Model 1.
[2)] Snow loads are only to be considered for covered bridges as an accompanying action.

Table 7: Standard values for deflections of road bridges

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21)	quasi-permanent (22)
Functionality – vertical relative displacement at expansion joints		$\delta_v \leq 5$ mm [1) 2) 3)]	
Comfort		$w \leq l/500$ [4)]	
Appearance			$w \leq l/700$ [1) 2)]

[1)] Deflection after deduction of hogging, if present. Any long-term effects due to shrinkage, relaxation or creep shall be considered.
[2)] Deflection due to the actions and long-term effects after installation of the relevant technical equipment.
[3)] If fittings or equipment are especially sensitive to structural deformations, besides or instead of dimensioning measures, above all constructional measures shall be implemented to prevent deterioration.
Guidelines of the producers and suppliers of the construction products to be fitted shall be observed.
[4)] Deflection due to the frequent value of Load Model 1.

The deflections shall be determined according to codes SIA 262 to 266.

Deviating limit values for deflections can be agreed upon in accordance with the service requirements and must be specified in the basis of design. In particular, for secondary structural members less stringent requirements can apply.

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APPENDIX C (normative)
PEDESTRIAN AND CYCLE-PATH BRIDGES

Table 8: Reduction factors for pedestrian and cycle-path bridges

Actions	ψ_0	ψ_1	ψ_2
Vertical loads			
– Load Model 1	0,4	0,4	0
– Load Model 2	0	0	0
Horizontal forces [1)]	0,4	0,4	0
Snow loads [2)]	0,6	0,2	0
Wind forces	0,6	0,2	0
Temperature actions	0,6	0,6	0,5
Actions imposed by the ground			
– earth pressure	0,7	0,7	0,7
– water pressure	0,7	0,7	0,7

1) The horizontal forces shall be assumed to act simultaneously with Load Model 1.
2) Snow loads are only to be considered for covered bridges as an accompanying action.

Table 9: Standard values for deflections of pedestrian and cycle-path bridges

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21)	quasi-permanent (22)
Functionality			
– deflection within span			$w \leq l/700$ [1) 2) 3)]
– vertical relative displacement at expansion joints		$\delta_v \leq 5$ mm [1) 2) 3)]	
Comfort		$w \leq l/600$ [4)]	
Appearance			$w \leq l/700$ [1) 2)]

1) Deflection after deduction of hogging, if present. Any long-term effects due to shrinkage, relaxation or creep shall be considered.
2) Deflection due to the actions and long-term effects after installation of the relevant technical equipment.
3) If fittings or equipment are especially sensitive to structural deformations, besides or instead of dimensioning measures, above all constructional measures shall be implemented to prevent deterioration.
Guidelines of the producers and suppliers of the construction products to be fitted shall be observed.
4) Deflection due to Load Model 1.

The deflections shall be determined according to codes SIA 262 to 266.

Deviating limit values for deflections can be agreed upon in accordance with the service requirements and must be specified in the basis of design. In particular, for secondary structural members less stringent requirements can apply.

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Table 10: Standard values for eigenfrequencies for pedestrian and cycle-path bridges

Limit state	Eigenfrequency [Hz]
Comfort – vertical vibrations – horizontal vibrations (transverse) – horizontal vibrations (longitudinal)	 $f > 4{,}5$ or $f < 1{,}6$ $f > 1{,}3$ $f > 2{,}5$
Eigenfrequencies deviating from the given standard values are admissible, provided a more exact dynamic analysis is carried out taking damping into consideration.	

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APPENDIX D (normative)
NORMAL GAUGE RAILWAY BRIDGES

Table 11: Reduction factors for normal gauge railway bridges

Actions	ψ_0	ψ_1	ψ_2
Vertical loads			
– Load Model 1	1,0	1,0 [1)]	0 [2)]
– Load Model 2	1,0	1,0 [1)]	0 [2)]
– Load Model 3	0	1,0	0 [3)]
Horizontal forces [4)]	1,0	1,0 [1)]	0
Wind forces			
– in general	0,8	0,5	0
– aerodynamic forces from rail operations	1,0	0,5	0
Temperature actions	0,6	0,6	0,5
Actions imposed by the ground			
– earth pressure	0,7	0,7	0,7
– water pressure	0,7	0,7	0,7

[1)] Applies to two loaded tracks.
[2)] In general $\psi_2 = 0$
For earthquake dimensioning situation $\psi_2 = 0{,}3$
For impact dimensioning situation $\psi_2 = 1{,}0$
[3)] In general $\psi_2 = 0$
For derailment dimensioning situations $\psi_2 = 1{,}0$
[4)] The horizontal forces shall be assumed to act as action groups simultaneously with the vertical loads.

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Table 12: Serviceability limits for normal gauge railway bridges

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21) [1]	quasi-permanent (22)
Functionality [2] – deflections [3] – $v \le 80$ km/h – 80 km/h $\le v \le$ 200 km/h – track twist [4] – $v \le 120$ km/h – 120 km/h < v ≤ 200 km/h – $v >$ 200 km/h – relative vertical displacement of deck ends behind abutments [5] – $v \le 160$ km/h – $v >$ 160 km/h		 $w \le l/800$ $w \le l/(15v-400)$ $\alpha_t \le 1{,}0$ mrad/m $\alpha_t \le 0{,}7$ mrad/m $\alpha_t \le 0{,}3$ mrad/m $\delta_v \le 3$ mm $\delta_v \le 2$ mm	
Appearance			$w \le l/700$ [6] [7]

[1] Deflection due to the variable leading action only shall be considered.

[2] For structures with multiple tracks, the functionality shall be verified for up to two simultaneously loaded tracks. The serviceability limits are valid for structures with ballasted track. For structures with slab track, the serviceability limits shall be specified after prior agreement with the supervisory authorities.

[3] Deflection due to Load Model 1 or 2, including factors Φ and α. The factor α may be taken as 1,0.

[4] Relevant track twist due to:
– Load Model 1 or 2 including factors Φ and α and associated horizontal forces or
– Load Model 3 including factor Φ and associated horizontal forces.

[5] Relative vertical displacement due to:
– Load Model 1 or 2 including factors Φ und α.

[6] Deflection after deduction of hogging, if present. Any long-term effects due to shrinkage, relaxation or creep shall be considered.

[7] Deflection due to the actions and long-term effects after installation of the relevant technical equipment.

The deformations shall be determined according to codes SIA 262 to 266.

Deviating serviceability limits can be determined after prior agreement with the supervisory authorities and shall be specified in the service criteria agreement. In particular, for secondary structural members and special sections such as feeder tracks and multiple track lines, less stringent requirements may apply.

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APPENDIX E (normative)
NARROW GAUGE RAILWAY BRIDGES

Table 13: Reduction factors for narrow gauge railway bridges

Actions	ψ_0	ψ_1	ψ_2
Vertical loads			
– Load Model 4	1,0	1,0 [1)]	0 [2)]
– Load Model 5	1,0	1,0 [1)]	0 [2)]
– Load Model 6	0	1,0	0
– Load Model 7	0	1,0	0
Horizontal forces [3)]	1,0	1,0 [1)]	0
Wind forces			
– in general	0,8	0,5	0
– aerodynamic forces from rail operations	1,0	0,5	0
Temperature actions	0,6	0,6	0,5
Actions imposed by the ground			
– earth pressure	0,7	0,7	0,7
– water pressure	0,7	0,7	0,7

[1)] Applies to two loaded tracks.

[2)] In general $\psi_2 = 0$
For earthquake dimensioning situation $\psi_2 = 0{,}3$
For impact dimensioning situation $\psi_2 = 1{,}0$

[3)] The horizontal forces shall be assumed to act as action groups simultaneously with the vertical loads.

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Table 14: Serviceability limits of narrow gauge railway bridges

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21) [1]	quasi-permanent (22)
functionality [2] – deflections [3] – $v \leq 80$ km/h – 80 km/h < $v \leq 120$ km/h – track twist [4] – relative vertical displacement of deck ends behind abutments [5]		 $w \leq l/800$ $w \leq l/(15v-400)$ $\alpha_t \leq 1{,}0$ mrad/m $\delta_v \leq 3$ mm	
Appearance			$w \leq l/700$ [6]

[1] Deflection due to the variable leading action only shall be considered.

[2] For structures with multiple tracks, the functionality shall be verified for up to two simultaneously loaded tracks. The serviceability limits are valid for structures with ballasted track. For structures with slab track, the serviceability limits shall be specified after prior agreement with the supervisory authorities.

[3] Deflection due to Load Model 4 or 5 and if applicable, due to Load Model 6 or 7, including factors Φ and α. The factor α may be taken as 1,0.

[4] Track twist due to horizontal forces and Load Model 4 or 5 and if applicable, due to Load Model 6 or 7, including factors Φ and α.

[5] Relative vertical displacement due to actions, which have to be considered for deflection.

[6] Deflection after deduction of hogging, if present. Any long-term effects due to shrinkage, relaxation or creep shall be considered.

The deformations shall be determined according to codes SIA 262 to 266.

Deviating serviceability limits can be determined after prior agreement with the supervisory authorities and shall be specified in the service criteria agreement. In particular, for secondary structural members and special sections such as feeder tracks and multiple track lines, less stringent requirements may apply.

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APPENDIX F (normative)
CRANE SUPPORTING STRUCTURES

Table 15: Reduction factors for crane supporting structures

Actions	ψ_0	ψ_1	ψ_2
Crane loads	1,0	1,0	0
Horizontal forces [1)]	1,0	1,0	0
Snow loads	$1 - 60/h_0$	$1 - 250/h_0$	$1 - 1000/h_0$
Wind forces	0,6	0,5	0
Temperature actions	0,6	0,5	0
Actions imposed by the ground – earth pressure – water pressure	 0,7 0,7	 0,7 0,7	 0,7 0,7
[1)] The horizontal forces shall be assumed to act simultaneously with the crane loads.			

Table 16: Standard value for deflection and displacements of crane supporting beams and columns

Limit state	Consequences of effects of actions		
	irreversible	reversible	reversible
	Load case		
	occasional (20)	frequent (21)	quasi-permanent (22)
Functionality [1) 2)] – use and operation, crane supporting beam – use and operation, crane supporting columns		 $w \le l/700$ [3)] $u \le l/800$ $u \le h/300$ $\Delta u \le 0{,}02$ m [4)]	
[1)] Due to variable actions. [2)] Crane loads without dynamic factor. [3)] For multiple cranes on one crane supporting structure, normally two independently moving cranes shall be considered. [4)] Difference in displacement of two columns facing each other, on the level of the crane rail.			
The horizontal displacements shall be determined according to codes SIA 262 to 266. Deviating limit values for horizontal displacements can be agreed upon in accordance with the service requirements and must be specified in the basis of design.			

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Abbreviations of the organisations represented in the Commission SIA 260

Empa	Swiss Federal Laboratories for Materials Science and Technology
EPFL	Swiss Federal Institute of Technology, Lausanne
FEDRO	Federal Roads Office
FOT	Federal Office of Transport

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Approval and validity

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It is valid as from 1st of August 2013.

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